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Учреждение образования
«Брестский Государственный технический университет»
Кафедра ИИТ

Лабораторная работа №8
По дисциплине «Мобильные приложения для информационных систем»

Выполнил:
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Брест 2026

Цель работы: получить практические знания работы с Android Studio.

Ход работы

GeoManager.java

```
package com.example.lab8new8;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;

import java.util.ArrayList;

public class GeoManager {

    private final GeoDatabase dbHelper;

    public GeoManager(Context context) {
        dbHelper = new GeoDatabase(context);
    }

    public void addLocation(double lat, double lng) {
        SQLiteDatabase db = dbHelper.getWritableDatabase();

        ContentValues values = new ContentValues();
        values.put("lat", lat);
        values.put("lng", lng);
        values.put("time", System.currentTimeMillis());

        db.insert("locations", null, values);
        db.close();
    }

    public ArrayList<LocationPoint> getAllPoints() {
        ArrayList<LocationPoint> list = new ArrayList<>();

        SQLiteDatabase db = dbHelper.getReadableDatabase();
        Cursor cursor = db.rawQuery("SELECT lat, lng, time FROM locations",
null);

        while (cursor.moveToNext()) {
            double lat = cursor.getDouble(0);
            double lng = cursor.getDouble(1);
            long time = cursor.getLong(2);

            list.add(new LocationPoint(lat, lng, time));
        }

        cursor.close();
        db.close();

        return list;
    }

    public ArrayList<LocationPoint> getPointsSince(long timeMillis) {
        ArrayList<LocationPoint> list = new ArrayList<>();

        SQLiteDatabase db = dbHelper.getReadableDatabase();
        Cursor cursor = db.rawQuery(
            "SELECT lat, lng, time FROM locations WHERE time >= ?",
            new String[]{String.valueOf(timeMillis)}
        );
    }
}
```

```

    );

    while (cursor.moveToNext()) {
        list.add(new LocationPoint(
            cursor.getDouble(0),
            cursor.getDouble(1),
            cursor.getLong(2)
        ));
    }

    cursor.close();
    db.close();

    return list;
}

public ArrayList<LocationPoint> getTodayPoints() {
    long now = System.currentTimeMillis();
    long dayAgo = now - 24L * 60 * 60 * 1000;
    return getPointsSince(dayAgo);
}

public ArrayList<LocationPoint> getWeekPoints() {
    long now = System.currentTimeMillis();
    long weekAgo = now - 7L * 24 * 60 * 60 * 1000;
    return getPointsSince(weekAgo);
}

public ArrayList<LocationPoint> getMonthPoints() {
    long now = System.currentTimeMillis();
    long monthAgo = now - 30L * 24 * 60 * 60 * 1000;
    return getPointsSince(monthAgo);
}

public double calculateTotalDistance(ArrayList<LocationPoint> points) {
    if (points.size() < 2) return 0;

    double total = 0;

    for (int i = 1; i < points.size(); i++) {
        LocationPoint p1 = points.get(i - 1);
        LocationPoint p2 = points.get(i);

        float[] result = new float[1];
        android.location.Location.distanceBetween(
            p1.lat, p1.lng,
            p2.lat, p2.lng,
            result
        );

        total += result[0]; // метры
    }

    return total; // метры
}

public double calculateAverageSpeed(ArrayList<LocationPoint> points) {
    if (points.size() < 2) return 0;

    long start = points.get(0).time;
    long end = points.get(points.size() - 1).time;

    double hours = (end - start) / 3600000.0;
    double km = calculateTotalDistance(points) / 1000.0;

    if (hours == 0) return 0;
}

```

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        return km / hours; // KM/ч
    }
    public ArrayList<String> getHistory() {
        ArrayList<String> list = new ArrayList<>();

        SQLiteDatabase db = dbHelper.getReadableDatabase();
        Cursor cursor = db.rawQuery("SELECT lat, lng, time FROM locations",
null);

        while (cursor.moveToNext()) {
            double lat = cursor.getDouble(0);
            double lng = cursor.getDouble(1);
            long time = cursor.getLong(2);

            String formatted = String.format(
                "Широта: %.5f\nДолгота: %.5f\nВремя: %s",
                lat, lng,
                new java.text.SimpleDateFormat("dd.MM.yyyy HH:mm:ss")
                    .format(new java.util.Date(time))
            );

            list.add(formatted);
        }

        cursor.close();
        db.close();

        return list;
    }
}

```

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```

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        values.put("lat", lat);
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        values.put("time", System.currentTimeMillis());

        db.insert("locations", null, values);
        db.close();
    }
}

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```

public ArrayList<LocationPoint> getAllPoints() {
    ArrayList<LocationPoint> list = new ArrayList<>();

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    Cursor cursor = db.rawQuery("SELECT lat, lng, time FROM locations",
null);

    while (cursor.moveToNext()) {
        double lat = cursor.getDouble(0);
        double lng = cursor.getDouble(1);
        long time = cursor.getLong(2);

        list.add(new LocationPoint(lat, lng, time));
    }

    cursor.close();
    db.close();

    return list;
}
public ArrayList<LocationPoint> getPointsSince(long timeMillis) {
    ArrayList<LocationPoint> list = new ArrayList<>();

    SQLiteDatabase db = dbHelper.getReadableDatabase();
    Cursor cursor = db.rawQuery(
        "SELECT lat, lng, time FROM locations WHERE time >= ?",
        new String[]{String.valueOf(timeMillis)}
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    while (cursor.moveToNext()) {
        list.add(new LocationPoint(
            cursor.getDouble(0),
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    }

    cursor.close();
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    long now = System.currentTimeMillis();
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    long now = System.currentTimeMillis();
    long weekAgo = now - 7L * 24 * 60 * 60 * 1000;
    return getPointsSince(weekAgo);
}

public ArrayList<LocationPoint> getMonthPoints() {
    long now = System.currentTimeMillis();
    long monthAgo = now - 30L * 24 * 60 * 60 * 1000;
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```

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            float[] result = new float[1];
            android.location.Location.distanceBetween(
                p1.lat, p1.lng,
                p2.lat, p2.lng,
                result
            );

            total += result[0]; // метры
        }

        return total; // метры
    }

    public double calculateAverageSpeed(ArrayList<LocationPoint> points) {
        if (points.size() < 2) return 0;

        long start = points.get(0).time;
        long end = points.get(points.size() - 1).time;

        double hours = (end - start) / 3600000.0;
        double km = calculateTotalDistance(points) / 1000.0;

        if (hours == 0) return 0;

        return km / hours; // км/ч
    }

    public ArrayList<String> getHistory() {
        ArrayList<String> list = new ArrayList<>();

        SQLiteDatabase db = dbHelper.getReadableDatabase();
        Cursor cursor = db.rawQuery("SELECT lat, lng, time FROM locations",
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                    .format(new java.util.Date(time))
            );

            list.add(formatted);
        }

        cursor.close();
        db.close();

        return list;
    }
}

```

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